

1a

$$\alpha = 0.98$$

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.98}{1-0.98}$$

$$\beta = 49$$

$$R_B = 100 R_C \text{ for } I_C \approx 100 I_B$$

Assume forward-active, $V_{BE} = 0.7V$

$$V_E = 0, \text{ so } V_B = 0.7$$

$$I_B = \frac{V_{CC} - V_B}{R_B} = \frac{5 - 0.7}{100k}$$

$$I_B = 0.043mA$$

$$I_C = \beta I_B = 49 \cdot 0.043mA$$

$$I_C = 2.107mA$$

$$I_C = I_S \left[e^{(V_{BE}/V_T)} - 1 \right] \text{ insignificant, ignore}$$

$$V_T \ln \frac{I_C}{I_S} = V_{BE}$$

$$V_{BE} = 0.026 \cdot \ln \left(\frac{2.107 \cdot 10^{-3}}{10^{-15}} \right)$$

$$I_C = \frac{5 - 0.738}{2041}$$

$$I_C = 2.09mA, \text{ small change, done}$$

$$V_{BE} = 0.738V$$

iterate

BE junction ON when $V_{BE} > 0.7V$

$$0.738V > 0.7V \checkmark$$

BC junction off when $V_{BC} < 0.5V$

$$V_B = 0.738V, \frac{V_{CC} - V_C}{R_C} = I_C$$

$$V_{BC} = 0.738 - 2.9$$

$$V_{BC} = -2.16V$$

$$-2.16V < 0.5V \checkmark$$

$$V_C = V_{CC} - I_C R_C$$

$$V_C = 5 - 2.1mA \cdot 1k$$

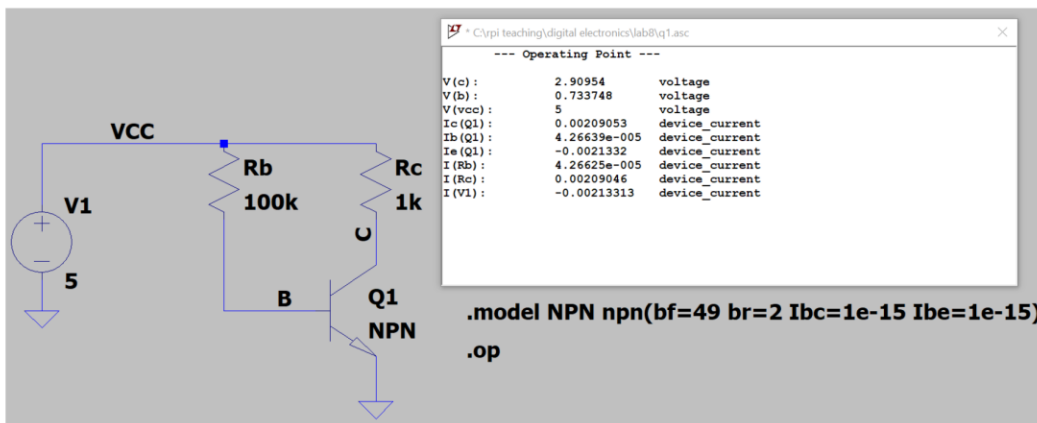
$$V_C = 2.9V$$

$$V_{CE} = V_C - V_E = 2.9 - 0$$

$$V_{CE} = 2.9V$$

BE is on, BC is off,

FA confirmed



I_C and V_{CE} match hand calculations exactly.

b) $R_B = 10k\Omega$. Assume increase in I_B results in saturation
 $V_{BE} = 0.7V$ (BC junction turns on).

$$I_C = \frac{V_{CC} - V_C}{R_C} = \frac{5 - 0.2}{1k}$$

$$I_C = 4.8mA$$

$$V_{BC} = 0.5V, \text{ so } V_{CE} = V_C - V_E$$

$$V_B - V_C = 0.5, V_B - V_E = 0.7$$

$$0.7 + V_E - V_C = 0.5 \quad V_B = 0.7 + V_E$$

$$V_C - V_E = 0.2$$

$$V_{CE} = 0.2$$

$$I_C = I_{SE} e^{V_{BE}/V_T} - I_{SC} e^{V_{CE}/V_T}$$

$$V_{BE} = V_T \ln \left(\frac{I_C}{I_{SE}} \right)$$

This term should be included due to saturation, but is excluded due to lack of lecture material.

$$V_{BE} = 0.026 \cdot \ln \left(\frac{4.8 \cdot 10^{-3}}{10^{-15}} \right)$$

$$V_{BE} = 0.759V$$

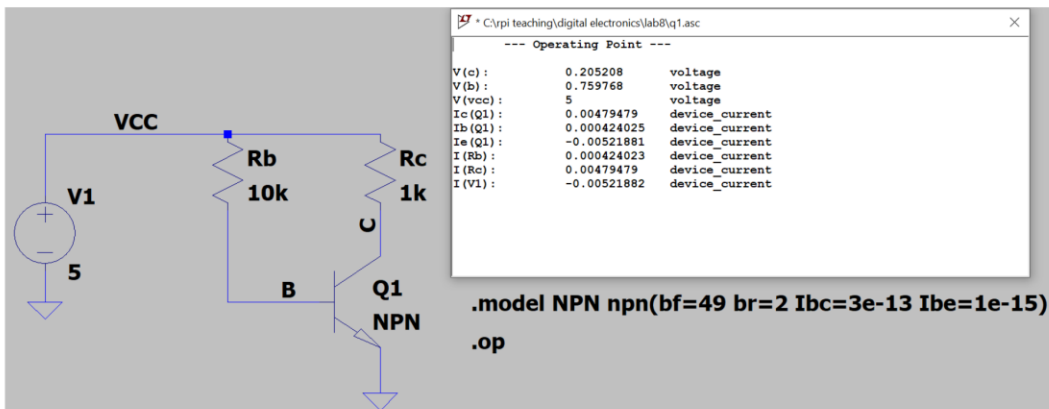
$I_C \neq \beta I_B$ in saturation, so no need to modify I_C

$$I_B = \frac{V_{CC} - V_B}{R_B} = \frac{5 - 0.7}{10k} = 0.43mA$$

Saturated if $I_C < \beta I_B$,

$$4.8mA < 49 \cdot 0.43mA$$

$$4.8mA < 21.07mA \checkmark$$



The given $V_{BE} = 0.7V$, $V_{BC} = 0.5V$, and resultant $V_{CE}(\text{sat}) = 0.2V$ assumed that BE junction reverse saturation current (I_{be} in SPICE) was much lower than the BC junction reverse saturation current (I_{bc} in SPICE), which is common for a real BJT. By default I_{be} and I_{bc} are equal in SPICE, resulting in a low V_{CE} of 0.056V. Multiplying I_{bc} by 300 results in the expected 0.2V V_{CE} . I_C is an exact match to hand calculations at 4.8mA.

c]

$$I_{s0} = \frac{A_E q D_n n_i^2}{N_E W_B} = 10^{-15} \text{ A}$$

$$\frac{I_{s1} = \frac{10 A_E q D_n n_i^2}{N_E W_B}}{I_{s0} = \frac{A_E q D_n n_i^2}{N_E W_B}} \quad I_{s1} = 10 I_{s0}, \quad \boxed{I_{s1} = 10^{-14} \text{ A}}$$

From part A, $V_{BE} = V_T \ln \left(\frac{I_C}{I_{sE}} \right)$

With $I_{sE1} = 10 I_{sE0}$, $V_{BE} = V_T \ln \left(\frac{I_C}{10 I_{sE}} \right) = V_T \left[\ln \frac{I_C}{I_{sE}} - \ln 10 \right]$

$$\Delta V_{BE} = -V_T \ln 10 = -0.026 \cdot 1.10$$

$$\boxed{\Delta V_{BE} = -0.06 \text{ V}}$$